

CODECHECK certificate 2026-026

<https://doi.org/10.5281/zenodo.23081945>



Amsterdam UMC

Item	Value
Title	Using an Open Science Checklist in Grant Proposal Reviews to Predict Reproducibility of Funded Publications
Authors	Ayu Putu Madri Dewi, Anne Floor Holtrop, Nicholas J DeVito, Melissa L. Rethlefsen, Zulfan Zazuli, Mathieu Repiquet, Ka Hin Tai, Constant Vinatier, Monika Varga, Stefan Wouters, Hengky Kurniawan, Nynke Swart, Gowri Gopalakrishna, Inge Stegeman, Mariska Leeftang
Ref. paper	https://osf.io/preprints/metaarxiv/dvhux_v1
Codechecker(s)	Merel Postema (ORCID: 0000-0002-1536-7062)
Date of Check	2026-07-28
Summary	Partial reproduction
Repository	https://github.com/codecheckers/funders

Table 1: CODECHECK summary

Summary

This codecheck resulted in a partial reproduction of the results, because I was unable to install the gt package in myDRE (Error: Chrome debugging port not open after 10 seconds). After several unsuccessful installation attempts, I proceeded without the package. As only the gtsave() function from gt was required, I was able to run the code by commenting out the gtsave() lines and saving the objects manually. However, this meant I could not create an interactive table and was limited to viewing only the first lines of the table outputs. These initial lines were reproducible, but I was unable to verify reproducibility for the entire table. The installation of this package remains a key challenge for any future reproduction.

Output	Comment
table_check.png	Table 1
table_control.png	Table 2
accuracy_table.png	Table 3
tg_funders.png	Annex 2, Table 1
tg_country.png	Annex 2, Table 2
tg_fied.png	Annex 2, Table 3
year_summary.jpeg	Annex 2, figure A
ROC_both.jpeg	Annex 2, figure B

Table 2: Summary of output files generated

CODECHECKER notes

No specialised computational environment was required to execute the code. However, at Amsterdam UMC, we use myDRE to conduct CODECHECKs. The clear instructions provided in the repository made it straightforward to reproduce the figures and tables, which could be generated by running a single script. Two minor modifications to the original code were necessary to ensure successful execution, as detailed below.

Installation prerequisites

The code was dependent on the following packages: readxl, pROC, dplyr, ggplot2, tools, openxlsx, tidyr, gtable, binom, gt, lattice, gtsummary, webshot2, stringr.

The code loaded the required packages without checking whether they were already installed. To improve reproducibility and ensure successful execution in a fresh environment, I added the following code to install any missing packages:

```
# Libraries
packages <- c(
  "readxl",
  "pROC",
  "dplyr",
  "ggplot2",
  "tools",
  "openxlsx",
  "tidyr",
  "gtable",
  "binom",
  "gt",
  "lattice",
  "gtsummary",
  "webshot2",
  "stringr"
)
install.packages(setdiff(packages, rownames(installed.packages())))
lapply(packages, library, character.only = TRUE)
```

Additionally, a line of code was added to load the stringr package, which was required for the code to run successfully.

Running the code

The code was executed within the Amsterdam UMC myDRE environment, which is specifically configured to support CODECHECKs. However, the protected environment did not permit a connection to Chrome, which is required by the gtsummary package. This initially resulted in the following error: **Error in initialize(...): Invalid path to Chrome**. To address this, the Renviron.site file was modified by adding the path to chrome.exe. Although this resolved the initial error, a subsequent error occurred that could not be resolved: **Error in startup(): ! Chrome debugging port not open after 10 seconds**. As a result, the lines containing the gtsave() function were commented out to allow the remainder of the code to run. The corresponding PNG files therefore had to be saved manually.

Manifest files

As noted above, the table outputs had to be saved manually due to issues with the gt package. Consequently, the tables could not be saved as interactive objects, preventing all rows from being inspected and the complete results from being verified. The results that could be checked were identical to those reported in the original manuscript, apart from minor formatting differences, such as unexpected symbols appearing where whitespace was present in the original preprint.

`table_check.png`

Comment: Table 1 in the manuscript.

2ÃŹ Table: Checklist Prediction vs Actual			
	Actual		Total
	Reproducible	Not reproducible	
Predicted			
Reproducible	2	15	17
Not reproducible	5	79	84
Total	7	94	101

table_control.png

Comment: Table 2 in the manuscript.

2ÃŹŹ Table: Control Prediction vs Actual			
	Actual		Total
	Reproducible	Not reproducible	
Predicted			
Reproducible	6	70	76
Not reproducible	1	24	25
Total	7	94	101

accuracy_table.png

Comment: Table 3 in the manuscript.

Accuracy Metrics per Group - Estimate (95% CI)		
Metric	Checklist	Control
Sensitivity	0.286 (0.082â“0.641)	0.857 (0.487â“0.974)
Specificity	0.840 (0.753â“0.901)	0.255 (0.178â“0.352)
PPV	0.118 (0.033â“0.343)	0.079 (0.037â“0.162)
NPV	0.940 (0.868â“0.974)	0.960 (0.805â“0.993)
Overall Accuracy	0.802 (0.714â“0.868)	0.297 (0.217â“0.392)

tg_funder.png

Comment: Annex 2, Table 1 in the manuscript.

Characteristic	N = 101 [†]
Funder Name	
Arkansas Audubon Society	1 (1.0%)
Australian Research Council	1 (1.0%)
BBSRC	2 (2.0%)
BELSPO	1 (1.0%)
Danish Research Council	1 (1.0%)
DFG	1 (1.0%)
DNSF	2 (2.0%)
Dresden Leibniz Graduate School	1 (1.0%)
European Union	9 (8.9%)
FWF	1 (1.0%)
GIDI	1 (1.0%)

tg_country.png

Comment: Annex 2, Table 2 in the manuscript.

Characteristic	N = 101 [†]
Country or Region	
Australia	2 (2.0%)
Austria	1 (1.0%)
Belgium	1 (1.0%)
Denmark	4 (4.0%)
EU	11 (11%)
Germany	3 (3.0%)
Greece	1 (1.0%)
Netherlands	4 (4.0%)
Romania	1 (1.0%)
Switzerland	1 (1.0%)
UK	3 (3.0%)

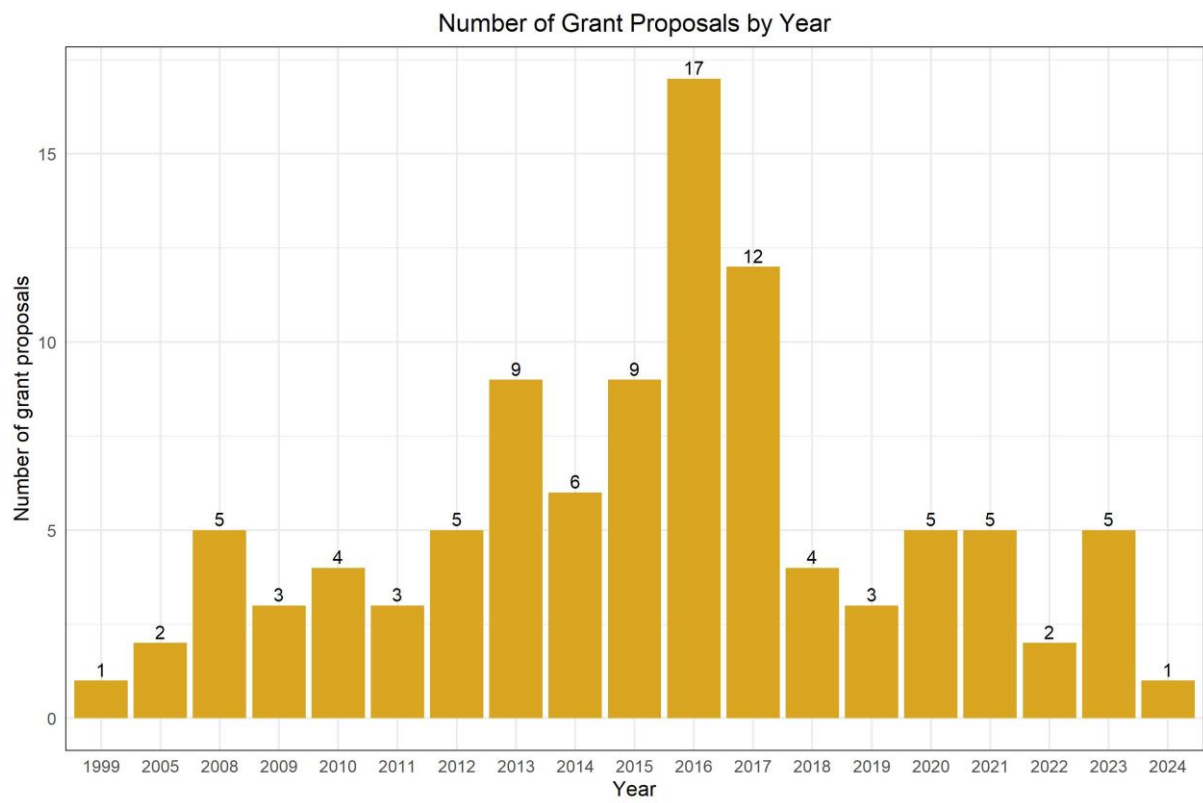
tg_field.png

Comment: Annex 2, Table 3 in the manuscript.

Characteristic	N = 101 [†]
Research Field (non-standardised category)	
Animal Science	1 (1.0%)
Anthropology	2 (2.0%)
Biochemistry	1 (1.0%)
Bioengineering	1 (1.0%)
Bioinformatics	4 (4.0%)
Biology	22 (22%)
Biomedicine	14 (14%)
Biophysics	1 (1.0%)
Biotechnology	1 (1.0%)
Computational Biology	1 (1.0%)
Computational Chemistry	3 (3.0%)

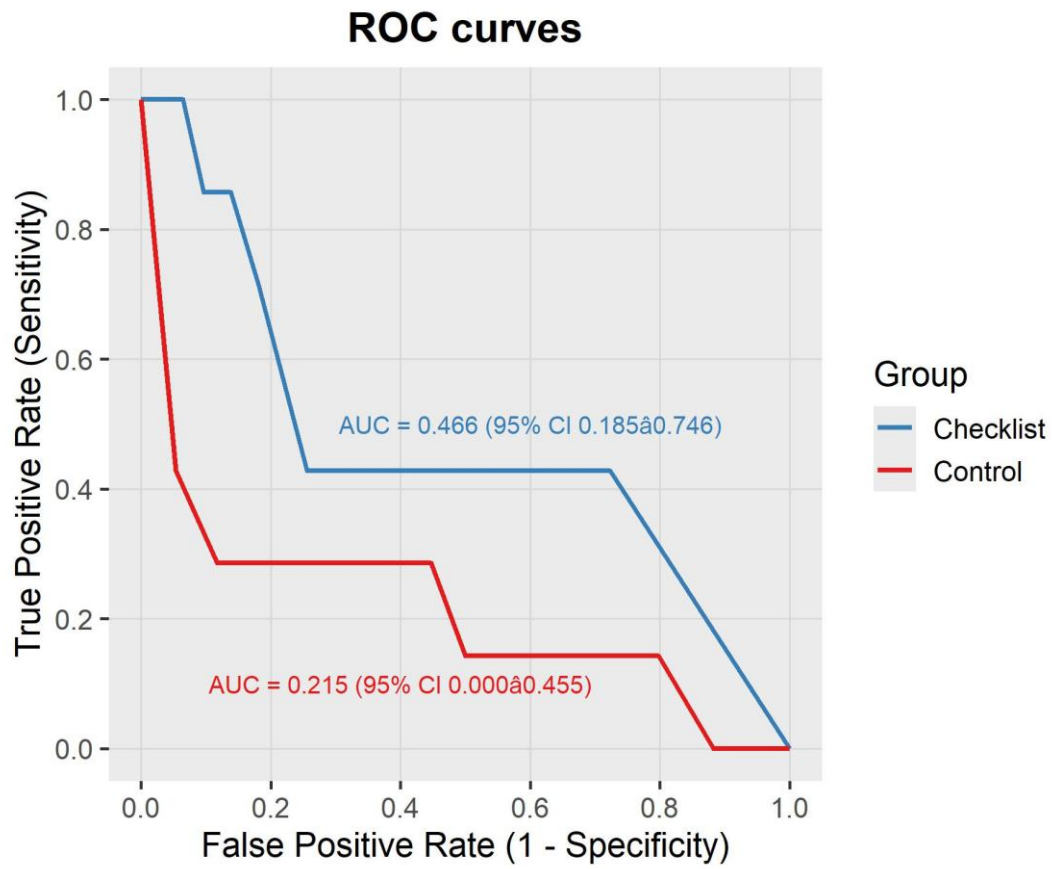
year_summary.jpeg

Comment: Annex 2, Figure A in the manuscript.



ROC_both.jpeg

Comment: Annex 2, Figure B in the manuscript.



Acknowledgements

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This CODECHECK was performed as part of an independent reproducibility review at the Research Data Management department, Amsterdam UMC, The Netherlands. The codechecker thanks the authors for making the code and data publicly available.

Citing this document

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